INTERNETWORKING DEVICES

Summarised Notes — Layer 1, Layer 2 & Layer 3 Connecting Devices

Contents

- **Layer 1 (Physical) connections** — Repeater, Hub
- **Layer 2 (Data Link) connections** — Bridge, Layer 2 Switch
- **Layer 3 (Network) connections** — Layer 3 Switch, Router, Gateway

1. Repeater

OSI Layer: Physical Layer (Layer 1)

Definition:

A repeater is a simple physical-layer networking device that receives a weak or distorted electrical/optical signal, regenerates and amplifies it, and retransmits it onto the next network segment. Its sole purpose is to extend the physical distance a signal can travel over a cable without loss of quality, since it works purely at the bit level and has no understanding of the data, addresses, or protocols it is carrying.

Advantages:

- Extends the maximum cable length / geographical range of a network.
- Simple, inexpensive, and easy to install.
- Regenerates both digital and analog signals, removing noise and distortion.
- Reduces signal attenuation over long distances.

Disadvantages:

- Does not filter traffic or reduce network congestion — simply repeats everything.
- Cannot connect networks of different types or speeds.
- Does not increase available bandwidth.
- Too many repeaters in series increase propagation delay.
- Provides no security or traffic-management features.

2. Hub

OSI Layer: Physical Layer (Layer 1)

Definition:

A hub is a basic multi-port physical-layer device that connects several devices in a network into a single segment. When it receives data on one port, it simply broadcasts (copies) that data out to every other connected port, regardless of the intended recipient; each device then decides for itself whether the frame was meant for it. Hubs may be passive (only join connections) or active (also regenerate/boost the signal like a multi-port repeater).

Advantages:

- Very simple, low-cost device — easy to set up.
- Useful for small networks or temporary/test setups.
- Can extend network reach like a multi-port repeater (active hubs).

Disadvantages:

- All connected devices share the same bandwidth, causing collisions and poor performance.
- Operates only in half-duplex mode.
- No filtering or intelligent forwarding — data is broadcast to all ports.
- No security features, since every device receives all traffic.
- Not suitable for large or high-traffic networks; mostly replaced by switches today.

3. Bridge

OSI Layer: Data Link Layer (Layer 2)

Definition:

A bridge is a networking device that connects two or more network segments of the same type (for example, LAN to LAN) at the Data Link layer of the OSI model. It learns the MAC addresses of devices on each connected segment and builds a MAC address table, which it then uses to filter and forward frames only to the segment where the destination device actually resides. This divides a large network into smaller collision domains, reducing unnecessary traffic and improving overall performance, while remaining transparent to the end users.

Advantages:

- Reduces network traffic and collisions by filtering frames based on MAC address.
- Divides a network into separate collision domains, improving efficiency.
- Extends the effective range of a LAN by joining segments.
- Transparent, plug-and-play operation for connected devices.

Disadvantages:

- Works only between networks of the same type (e.g., cannot join Ethernet to Token Ring).
- Cannot route data between different network-layer protocols.
- Does not understand IP addresses — limited to Layer 2 forwarding.
- Broadcasts and unknown-destination frames are still forwarded to all segments.
- Does not break up broadcast domains, only collision domains.

4. Layer 2 Switch

OSI Layer: Data Link Layer (Layer 2)

Definition:

A Layer 2 switch is essentially an intelligent, multi-port bridge that operates at the Data Link layer and uses MAC addresses to forward data efficiently within a LAN. It learns the MAC address of every device attached to it and builds a MAC address table mapping addresses to ports. When a frame arrives, the switch checks this table and forwards the frame only to the specific port where the destination device is located (or floods it if the address is unknown), instead of broadcasting to every port as a hub would. This gives each port its own collision domain and greatly improves performance and security, and many switches also support VLANs and the Spanning Tree Protocol (STP) to prevent loops.

Advantages:

- High performance and low latency due to store-and-forward, non-blocking switching.
- Reduces network congestion by sending frames only to the intended port.

- Each port is a separate collision domain, minimising collisions.
- Supports VLANs (802.1Q) for logical network segmentation and improved security.
- Supports Spanning Tree Protocol (STP/RSTP/MSTP) to prevent loops and broadcast storms.
- Scalable and cost-effective for enterprise, campus, and office LANs.

Disadvantages:

- Cannot route packets between different networks or subnets on its own.
- Works only on MAC addresses, not IP addresses — no Layer 3 awareness.
- All ports normally share the same broadcast domain unless VLANs are configured.
- Requires an external router for inter-VLAN or inter-network communication.

5. Layer 3 Switch

OSI Layer: Network Layer (Layer 3)

Definition:

A Layer 3 switch combines the high-speed, hardware-based frame switching of a Layer 2 switch with the IP-based routing intelligence of a router. It can perform inter-VLAN routing and route traffic between different IP subnets at near wire-speed, using dedicated routing hardware (ASICs) rather than software-based routing. In addition to switching, it supports VLAN interfaces (SVIs), access control lists (ACLs), DHCP relay, QoS, and other advanced management features, making it ideal for large enterprise, campus, and data-centre networks that need both speed and routing capability in a single device.

Advantages:

- Combines switching and routing in one device, reducing the need for a separate router.
- Very high performance — hardware-based routing gives low latency and high throughput.
- Simplifies network design through centralised inter-VLAN routing.
- Enhanced security via ACLs, IP source guard, and policy-based routing.
- Highly scalable for networks with many VLANs and subnets.
- Cost-effective in the long run — cuts hardware, power, and operational costs.

Disadvantages:

- More expensive than a basic Layer 2 switch (moderate-to-high cost).
- More complex to configure and manage, requiring networking expertise.
- Primarily suited to LAN/inter-VLAN routing — not a full replacement for WAN routers.
- Overkill (and unnecessarily costly) for small, simple networks.

6. Router

OSI Layer: Network Layer (Layer 3)

Definition:

A router is an intelligent Layer 3 device that connects two or more different networks (for example, a LAN to a WAN or the Internet) and forwards data packets between them based on IP addresses. It examines the destination IP address of each incoming packet, consults its routing table, and selects the best available path to forward the packet toward its destination. Routers also separate broadcast domains, exchange routing information with other routers using routing protocols, and commonly provide additional security features such as firewalling, NAT (Network Address Translation), and

access control.

Advantages:

- Connects different networks and enables communication between them (e.g., LAN to Internet).
- Chooses the optimal path for data, improving efficiency of delivery.
- Separates broadcast domains, reducing unnecessary network traffic.
- Provides built-in security features such as firewall, NAT, and access control.
- Supports multiple routing protocols for dynamic, scalable network design.

Disadvantages:

- More expensive and complex to configure than switches, bridges, or hubs.
- Routing decisions (especially software-based) can add latency compared to pure switching.
- Requires ongoing management of routing tables/protocols for optimal performance.
- A single point of failure for connected networks unless redundancy is configured.

7. Gateway

OSI Layer: Works across multiple OSI layers

Definition:

A gateway is a networking device (or software system) that connects two dissimilar networks which use different communication protocols or data formats, acting as a translator between them. Unlike a router, which simply forwards packets between networks using the same protocol suite, a gateway can convert data formats and protocols so that systems built on entirely different architectures can understand each other — for example, connecting a local network or IoT devices to cloud services, or linking a LAN to the Internet through an ISP modem. Because of this, a gateway is often described as acting like a combined bridge and translator between networks.

Advantages:

- Enables communication between networks using completely different protocols or formats.
- Provides a single access point connecting a local network to the Internet or cloud.
- Essential for IoT devices and diverse systems that need to reach cloud services.
- Works across multiple OSI layers, offering flexible protocol/format conversion.
- Widely used in enterprise networks and data centres to link dissimilar systems securely.

Disadvantages:

- Protocol translation adds processing overhead, which can introduce latency.
- More complex to configure and maintain than simple routers or switches.
- Can become a performance bottleneck or single point of failure for the networks it joins.
- Higher cost, especially for gateways handling multiple protocol conversions.

Quick Comparison Table

Device	OSI Layer	Address Used	Key Function
Repeater	Layer 1	None	Regenerates & amplifies signal
Hub	Layer 1	None	Broadcasts data to all ports
Bridge	Layer 2	MAC	Connects 2 segments, filters by MAC
Layer 2 Switch	Layer 2	MAC	Multi-port intelligent bridging
Layer 3 Switch	Layer 3	MAC + IP	Switching + inter-VLAN routing
Router	Layer 3	IP	Connects different networks, routing
Gateway	Multiple layers	Protocol-dependent	Translates between different protocols

Key takeaway: As we move up the OSI layers — from repeaters/hubs (Layer 1) to bridges/switches (Layer 2) to switches/routers/gateways (Layer 3 and above) — internetworking devices become progressively more 'intelligent', filtering and directing traffic based on increasingly specific information (signal → MAC address → IP address → protocol), which improves performance, scalability, and security at the cost of greater complexity and price.